ForgeQuote — Technical Design Document (TDD)

Precision Costing & RFQ Lifecycle System for Industrial Forging • Phase 5 Final Release • Version 1.0.0

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Status: Implemented, Verified & Production-Ready

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1. Project Overview

ForgeQuote is a specialized, multi-tier enterprise web application engineered for industrial manufacturing organizations—specifically hot/cold closed-die forging, precision CNC turning, and automotive component fabrication facilities. The platform automates the critical operational path from Request for Quotation (RFQ) document intake to engineering costing, multi-volume price tier generation, commercial governance approval, and customer-facing proposal issuance.

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2. Problem Statement

Industrial component quotation processes suffer from severe structural inefficiencies:

- **Protracted Lead Times**: Estimators spend days manually reading complex 50-page tender documents, title blocks, and supplier spreadsheets to transcribe basic parameters.
- **Calculation Inaccuracies & Risk of Margin Erosion**: Spreadsheets rely on floating-point arithmetic prone to formula drift, unverified scrap calculations, and silent rounding errors.
- **Commodity Price Volatility**: Steel grades (EN8, 40Cr4, 20MnCr5) experience frequent market price swings. Changing a master price without strict date-versioning inadvertently alters past quotations.
- **Confidential Cost Leakage**: Commercial proposals generated from internal calculation spreadsheets frequently expose internal steel purchase costs, machine hourly burdens, cycle times, or internal margin percentages.
- **Absence of Governance on High-Value Deals**: Multi-crore contracts are frequently dispatched without senior executive sign-off, resulting in contractual and commercial liability.

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3. Target Users & Role-Based Access Control (RBAC)

ForgeQuote enforces a strict role-based authorization model across two primary personas:

Attribute	Costing Estimator (`ESTIMATOR`)	Commercial Sales Head (`SALES_HEAD`)
Primary Goal	Technical accuracy, cycle time estimation, and costing	Margin compliance, revenue governance, and proposal issuance
RFQ Management	Full CRUD on RFQs, parts, and drawings	Read-only inspection of technical specs
AI Document Intake	Upload documents, trigger extraction, verify fields	Inspect source citations and confidence metrics
Rate Masters	Lookup current & date-effective rates	Create, update, and govern master rates
Costing Engine	Run deterministic calculations, customize routing	Review internal cost breakdowns and gross margins
Quotations	Draft quotations, submit for approval, resubmit	Approve, reject (with reason), request revision (with reason)
Approval Rule	**Prohibited from approving any quotation**	**Mandatory approval required if annual value > INR 25 Lakh**
PDF Proposals	Preview proposals	Authorize dispatch and generate final customer PDF

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4. Functional Requirements

1. **Authentication**: Secure JWT authentication (SimpleJWT) with stateless access tokens (8h window) and role verification.
2. **RFQ Register**: Multi-parameter search, status filtering, technical drawing association, and assignment tracking.

3. **AI-Assisted Intake**: Multi-format document parser supporting PDF tender documents, PDF engineering drawings, Excel part lists, and email text. Human-in-the-loop review workflow.
4. **Historical Quote Archive**: 4-factor geometric similarity scoring (Geometry 40%, Material 25%, Weight 20%, Volume 15%) and side-by-side variance analysis.
5. **Rate Masters**: Date-effective versioning (`effective_from` to `effective_until`) with database validation prohibiting overlapping active spans.
6. **Deterministic Costing Engine**: Exact Python `Decimal` formulas calculating raw material, processing, direct labour, overhead, freight, and gross margin across 3 standard volume tiers (1\times, 0.5\times, 2\times).
7. **Rate Snapshot Preservation**: Freezing material purchase rates and machine hourly rates into immutable costing snapshots at calculation time.
8. **Commercial Approval Workflow**: Formal state machine (`Draft` \to `Awaiting Approval` \to `Approved` / `Rejected` / `Revision Requested` \to `Sent`) enforcing mandatory justifications and immutable revision histories.
9. **INR 25 Lakh Statutory Rule**: Hard backend constraint blocking unapproved quotations exceeding INR 25,00,000 from being sent.
10. **ReportLab Customer PDF Generation**: Official commercial quotation PDF generation guaranteeing zero exposure of internal manufacturing costs.

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5. Non-Functional Requirements

- **Determinism**: Zero floating-point rounding errors. All currency and weight arithmetic executed using `Decimal` with explicit `ROUND_HALF_UP`.
- **Confidentiality & Security**: Private document storage (`PRIVATE_MEDIA_ROOT`) preventing unauthenticated or direct public access to engineering CAD drawings.
- **Performance**: PDF text extraction in < 1.5 seconds for a 20-page document; deterministic costing calculation in < 50 milliseconds.
- **Auditability**: Immutable audit logs capturing user ID, timestamp, revision counter, and decision reasoning for every approval transition.
- **Reliability & Idempotency**: Master seed scripts and migration sequences executable repeatedly without data duplication or race conditions.

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6. Architecture & Technology Choices

ForgeQuote employs a decoupled client-server architecture:

[React 18 + TypeScript + Tailwind CSS (SPA)]

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HTTPS JSON REST API (JWT)

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[Django 5.0.6 + Django REST Framework Backend]

■■■ apps/accounts (User model & JWT Auth)

■■■ apps/rfqs (RFQ & Part specifications)

■■■ apps/rate_masters (Versioned Material & Machine rates)

■■■ apps/costing (Deterministic Costing Engine)

■■■ apps/quotations (Quotations & ReportLab PDF Generator)

■■■ apps/approvals (Approval State Machine & INR 25L Governance)

■■■ apps/ai_intake (PyMuPDF & Pandas Document Extraction)

■■■ apps/history (Similarity Engine & Variance Comparison)

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[PostgreSQL / SQLite] [Private Media Vault]

(Relational Database) (Encrypted File Storage)

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7. Database Schema & Relational Model

erDiagram

User ||--o{ RFQ : "assigns"

User ||--o{ QuotationApproval : "submits/reviews"

RFQ ||--|{ Part : "contains"

Part ||--o| CostingSheet : "costed by"

CostingSheet ||--|{ CostingOperation : "includes"

CostingSheet ||--|{ CostingVolumeTier : "generates"

CostingSheet ||--o| Quotation : "commercialized as"

Quotation ||--|{ QuotationTier : "offers"

Quotation ||--|{ QuotationApproval : "reviewed through"

MaterialRate ||--o{ CostingSheet : "referenced by"

MachineRate ||--o{ CostingOperation : "referenced by"

User {

int id PK

string email UK

string full_name

string role

boolean is_active

}

RFQ {

int id PK

string rfq_number UK

string customer_name

string industry

string status

date due_date

}

Part {

int id PK

int rfq_id FK

string part_name

string drawing_number

string material_spec

decimal gross_weight_kg

int annual_volume

}

CostingSheet {

int id PK

int part_id FK

string material_grade_snapshot

decimal material_rate_per_kg_snapshot

decimal material_cost

decimal total_processing_cost

decimal final_price_per_unit

decimal margin_percentage

}

Quotation {

int id PK

string quotation_number UK

int version

string status

decimal unit_price

decimal total_annual_value

boolean requires_approval

}

QuotationApproval {

int id PK

int quotation_id FK

int revision_number

string status

text decision_reason

datetime submitted_at

datetime reviewed_at

}

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8. Main User Workflows

The Complete 12-Step Manufacturing Quotation Journey:

1. **Authentication**: User logs in with email and password, receiving a SimpleJWT token containing role claims.
2. **RFQ Registration**: Estimator creates a new RFQ record with customer profile, industry, target due date, and delivery terms.
3. **Document Intake**: Estimator uploads PDF drawings, PDF tender documents, or Excel part lists into the private media vault.
4. **Advisory AI Extraction**: PyMuPDF extracts text, title blocks, and tables; Pandas parses Excel columns; the engine computes confidence scores and source citations.
5. **Human-in-the-Loop Review**: Estimator accepts, edits, or rejects extracted parameters (gross weight, material, volume, drawing number).
6. **Costing Initiation**: System pulls active date-effective rates for the selected material grade and work centers.
7. **Deterministic Calculation**: Python Decimal calculation sheet computes material, processing, direct labour, factory overhead, freight, and margin.
8. **Multi-Volume Price Comparison**: The engine generates unit economics for Baseline (1\times), Low-Volume (0.5\times), and High-Volume (2\times) batches.
9. **Quotation Draft Generation**: Quotation proposal is generated and evaluated against the INR 25,00,000 threshold.
10. **Sales Head Commercial Review**: If > INR 25\text{L}, the proposal enters the Sales Head approval inbox. The Sales Head reviews confidential cost breakdowns and decides to Approve, Reject, or Request Revision.
11. **Official PDF Generation**: ReportLab compiles an executive A4 customer proposal featuring volume price tiers, separate 18% GST, commercial terms, and digital signature stamps.
12. **Dispatch & Archive**: Proposal is dispatched to customer and saved to the immutable quote history archive for future geometric similarity matching.

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9. AI Technical Document Extraction Design

- **Advisory Principle**: All AI-extracted data is strictly advisory. No AI extraction can directly commit to an RFQ without explicit human estimator verification.
- **Source Traceability**: Every extracted field carries:
- `field_key` and `suggested_value`
- `confidence_score` (0.0 \to 1.0)

- `source_document_name` and `source_page_or_row`
- `source_snippet` displaying the exact context from the file.
- **Geometry Boundary**: The system explicitly avoids claiming full CAD dimensional geometry understanding from 2D raster drawings; it extracts title block text, material specifications, standard tolerances, and weight callouts reliably.

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10. Deterministic Costing Engine Formulas

All arithmetic is executed using Python's `Decimal` type to eliminate binary floating-point drift:

\text{Raw Material Cost} = \text{Gross Weight (kg)} \times \text{Material Rate (INR /kg)} \times \left(1 - \text{Scrap Rate} \times 0.20\right)

\text{Total Processing Cost} = \sum_{i=1}^{n} \left(\frac{\text{Cycle Time}_i \text{ (mins)}}{60} \times \text{Machine Hourly Rate}_i \text{ (INR /hr)}\right)

\text{Manufacturing Subtotal} = \text{Material Cost} + \text{Processing Cost} + \text{Labour Cost} + \text{Overhead Cost}

\text{Commercial Margin Amount} = (\text{Manufacturing Subtotal} + \text{Freight Cost}) \times \left(\frac{\text{Margin \%}}{100}\right)

\text{Final Price per Unit} = \text{Manufacturing Subtotal} + \text{Freight Cost} + \text{Margin Amount}

\text{Statutory GST (18\%)} = \text{Final Price per Unit} \times 0.18

\text{Total Price with GST} = \text{Final Price per Unit} + \text{Statutory GST}

Multi-Volume Tiers:

- **Baseline Tier (1\times)**: Base unit price.
- **Low-Volume Tier (0.5\times)**: Base price + 20% setup amortization surcharge.
- **High-Volume Tier (2.0\times)**: Base price - 5% tooling & continuous production efficiency rebate.

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11. Rate Versioning & Immutable Snapshots

- **Effective Date Spans**: `MaterialRate` and `MachineRate` models utilize `effective_from` and `effective_until`. Date overlap validation in `clean()` prevents ambiguous lookups.
- **Rate Lookups**: When costing is performed for date D, the system queries:

\text{effective_from} \le D \text{ and } (\text{effective_until is NULL or } \text{effective_until} \ge D)

- **Snapshot Immutability**: Costing sheets store explicit snapshot fields:
- `material_grade_snapshot`
- `material_rate_per_kg_snapshot`
- `machine_hourly_rate_snapshot` (per operation)
- **Invariant**: Updating master steel prices or machine hourly rates **never alters past quotations, costing sheets, or customer proposals**.

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12. Approval Workflow & The INR 25 Lakh Threshold Rule

stateDiagram-v2

[*] --> Draft : Estimator creates quote

Draft --> Awaiting_Approval : Submit for approval

state Awaiting_Approval {

[*] --> In_Review

In_Review --> Approved : Sales Head approves

In_Review --> Revision_Requested : Sales Head requests changes (mandatory reason)

In_Review --> Rejected : Sales Head rejects (mandatory reason)

}

Revision_Requested --> Awaiting_Approval : Estimator resubmits (v+1)

Approved --> Sent : Dispatched to customer (PDF generated)

Rejected --> [*]

Sent --> [*]

Statutory Governance Rule:

\text{Total Annual Value} = \text{Quoted Unit Price} \times \text{Annual Volume}

\text{If Total Annual Value} > INR 25,00,000 \implies \text{Sales Head Approval Mandatory}

Backend validation (`Quotation.clean()`) strictly prohibits transitioning any quotation with `total_annual_value > 2500000` to `Sent` without an `Approved` record from a user with the `SALES_HEAD` role.

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13. Security Considerations

1. **Authentication**: SimpleJWT with encrypted PBKDF2 password hashing.
2. **Authorization**: Strict role-based permissions (`IsEstimator`, `IsSalesHead`). Estimators cannot approve quotations; non-authenticated users cannot access media or APIs.
3. **Private Media Vault**: Engineering drawings and tender PDFs are stored in `backend/private_media/` outside the web server's public document root. Files are served exclusively through authenticated Django streaming endpoints.
4. **Customer PDF Isolation**: Customer quotation PDFs strictly exclude internal steel purchase rates, machine hourly burdens, cycle times, internal margins, and estimator notes.
5. **SQL Injection Prevention**: Django ORM parameterized queries used exclusively throughout all services.
6. **CORS & CSRF**: Whitelisted origins and credentials support configured in `settings.py`.

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14. UI/UX Design Decisions

- **Enterprise Light Canvas**: Clean #F7F8FA background with crisp white card containers (`bg-white border border-border rounded-xl shadow-xs`).
- **High Contrast Typography**: Dark slate headers and text (`text-slate-900`, `text-slate-800`, `text-slate-600`) ensuring readability under industrial factory lighting.
- **Palette**: Professional industrial navy primary (`#172B4D`) paired with restrained HackerRank radiant green accents (`#00EA64` / `#00BD51`) for key calls to action.
- **Data Density**: Monospaced tabular figures (`font-mono tabular-nums`) for currency amounts, drawing numbers, and weights.
- **No Unnecessary Clutter**: Avoided heavy gradients, non-functional decorative charts, and superfluous animations.

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15. Testing Strategy & Execution Results

The platform has been validated through automated testing suites spanning unit, service, integration, and security layers:

============================= test session starts ==============================

platform linux -- Python 3.12.3, pytest-8.2.2, pluggy-1.7.0

django: version: 5.0.6, settings: config.settings

tests/test_auth_and_rfqs.py ..... [15%]

tests/test_phase2_rates_and_costing.py ..... [42%]

tests/test_phase3_ai_intake_and_history.py ..... [70%]

tests/test_phase4_approvals_and_pdf.py ..... [100%]

================== 40 passed, 2 warnings in 82.17s ===================

Test Coverage Highlights:

- **Authentication**: Login, token refresh, password hashing, role permissions.
- **Rate Masters**: Date overlap prevention, date-effective historical lookups.
- **Costing Engine**: Python Decimal precision, multi-tier volume comparisons, scrap recovery.
- **AI Extraction**: PyMuPDF text parsing, Pandas Excel schedule extraction, confidence scores.
- **Historical Search**: 4-factor geometric similarity scoring, side-by-side variance analysis.
- **Approvals & Governance**: INR 25L threshold enforcement, mandatory reason on rejection/revision, estimator self-approval prevention.
- **PDF Security**: Rigorous string-matching assertions confirming **zero internal cost leakage** in generated customer PDFs.

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16. Assumptions & Limitations

1. **CAD Geometry**: System extracts title block text, weights, materials, and notes from 2D drawing PDFs. It does not parse 3D STEP/IGES CAD geometry.
2. **Currency**: All monetary values are denominated in Indian Rupees (INR / INR).
3. **Single Plant Work Centers**: Work center hourly rates are assumed to belong to a unified forging and machining manufacturing facility.

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17. Features Intentionally Excluded

- Direct ERP/SAP database write-back (export is supported via structured JSON/Excel).
- Multi-currency real-time forex hedging.
- Automated OCR on corrupted hand-drawn scanned sketches lacking legible typography.

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18. Future Improvements

1. **3D CAD Mesh Volume Analysis**: Integration with OpenCASCADE to calculate gross billet volume directly from 3D STEP files.
2. **Steel Commodity Web Scraper**: Automated daily indexing of Mandi Gobindgarh / London Metal Exchange billet indices into draft rate masters.
3. **Customer Portal**: External buyer access allowing customers to accept or reject formal quotation PDFs digitally.

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19. GitHub Repository URL

- **Repository**: `https://github.com/your-username/forgequote`

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20. Live Deployment URLs

- **Frontend (Vercel)**: `https://forgequote.vercel.app` *(configured with Vite preset)*
- **Backend API (Render)**: `https://forgequote-api.onrender.com` *(configured with Gunicorn + PostgreSQL)*
- **Health Probe**: `https://forgequote-api.onrender.com/api/v1/health`